

# BRIAN POWELL

## Senior AI/ML Software Engineer

A: Navarre, FL 32566 P: (850) 236 2147 E: bripowell95@hotmail.com L: linkedin.com/in/brian-powell-03997b3b1 G: github.com/brian-powell

### SUMMARY

AI System Architect with 10+ years of experience building revenue-driving, production-scale AI systems across real estate, media intelligence, and energy infrastructure. Expert in LLM-powered architectures, RAG systems, advanced NLP, and large-scale predictive modeling, delivering high-accuracy, low-latency intelligence platforms on AWS using cloud-native MLOps and distributed computing. Proven track record of owning the full AI lifecycle end to end—from data engineering and model design to scalable microservices, automation pipelines, and real-time analytics—transforming complex, multi-source data into strategic, decision-grade insights.

### CORE COMPETENCIES

- End-to-End AI System Architecture
- Machine Learning & Deep Learning Engineering
- LLM & Retrieval-Augmented Generation Systems
- Large-Scale Data Engineering & Distributed Processing
- MLOps & Production Model Lifecycle Management
- High-Performance Backend & API Development
- Cloud-Native Infrastructure & DevOps Automation
- Real-Time Analytics & Interactive Data Visualization

### WORK EXPERIENCE

Nov 2023 – Dec 2025

#### Senior AI/ML Software Engineer

Blue Label Labs, New York, NY

- Architected an AI-powered real estate due diligence platform using **C++**, **Python 3.11**, **FastAPI 0.110**, and **AWS**, delivering structured feasibility and compliance reports from multi-source geospatial data in minutes.
- Designed a production-grade RAG system with **OpenAI GPT-4o**, **LangChain 0.1**, **LlamaIndex 0.10**, and **Milvus 2.3**, enabling property-level regulatory reasoning and contextual document intelligence.
- Built ML-driven document and zoning classification pipelines using **PyTorch 2.2**, **Hugging Face Transformers 4.38**, **XGBoost 2.0**, and **scikit-learn 1.4**, generating structured compliance and risk signals.
- Engineered large-scale dataset processing workflows with **Apache Spark 3.5**, **Apache Airflow 2.8**, **Pandas 2.2**, and **NumPy 1.26**, transforming ArcGIS layers and municipal bylaws into embedding-ready vector representations.
- Developed backend microservices and vector retrieval APIs using **PostgreSQL 15**, **Redis 7**, **SQLAlchemy 2.0**, and **gRPC**, supporting high-concurrency semantic search and low-latency regulatory querying.
- Deployed scalable cloud infrastructure and MLOps pipelines via **Amazon EKS**, **SageMaker**, **Terraform 1.6**, **GitHub Actions**, **MLflow 2.9**, and **Docker 25**, enabling automated model training, versioning, and CI/CD-driven container releases.
- Enabled real-time database-to-visualization synchronization using **React 18**, **WebSockets**, **Plotly Dash 2.14**, and **Amazon CloudFront**, ensuring low-latency interactive geospatial dashboards.

#### AI/ML Software Engineer

May 2019 – Oct 2023

Plego, Chicago, IL

- Architected automated news intelligence platform using **Python 3.10**, **Django 4.1**, and **Azure Functions**, delivering large-scale article ingestion and structured PR analytics reports.
- Designed NLP-driven content understanding pipelines with **spaCy 3.4**, **Gensim 4.3**, and **BERT 1.0**, enabling entity extraction, topic modeling, and semantic similarity scoring.
- Built classification and clustering models using **LightGBM 3.3**, **CatBoost 1.1**, and **TensorFlow 2.10**, generating automated article tagging, tier detection, and duplicate grouping.
- Engineered distributed dataset processing workflows with **Dask 2023.3**, **BeautifulSoup 4.12**, and **Celery 5.2**, executing large-scale scraping, parsing, and asynchronous task orchestration.
- Developed scalable backend services and high-performance APIs using **Oracle 19c**, **Elasticsearch 8.5**, and **gRPC 1.54**, enabling indexed retrieval and low-latency inter-service communication.
- Deployed containerized ML workloads on **Azure Container Apps**, **Azure Blob Storage**, and **Azure DevOps Pipelines**, implementing automated build, artifact storage, and CI/CD-driven releases.
- Enabled real-time dataset-to-dashboard synchronization using **Next.js 13**, **D3.js 7**, and **Apache Kafka 3.4**, delivering interactive visual analytics and export-ready reporting views.

## Data Scientist

Jun 2017 – Apr 2019

Electric Power Engineers, Austin, TX

- Developed short and long term load forecasting models using **Python**, **Pandas**, **Scikit-learn**, and **XGBoost**, enabling grid demand prediction and capacity planning optimization.
- Engineered large scale smart meter and SCADA data pipelines with **SQL**, **Apache Spark**, and **AWS S3**, transforming raw telemetry into structured analytical datasets for modeling.
- Built anomaly detection and predictive maintenance systems using **PyTorch** and **TensorFlow**, identifying transformer and substation fault risks before operational failure.
- Deployed real time monitoring APIs with **FastAPI**, containerized via **Docker**, orchestrated using **Kubernetes**, and tracked experiments through **MLflow** to ensure scalable and production ready energy AI systems.

## SKILLS

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**Programming Languages:** Python, JavaScript, TypeScript, C++, SQL

**AI / Machine Learning:** Scikit-learn, XGBoost, LightGBM, CatBoost, TensorFlow, PyTorch, spaCy, Gensim, Hugging Face Transformers, BERT

**LLM & Retrieval Systems:** OpenAI GPT-4o, LangChain, LlamaIndex, Milvus

**Data Processing & Engineering:** Apache Spark, Apache Airflow, Dask, Pandas, NumPy, BeautifulSoup, Celery

**Backend & APIs:** FastAPI, Django, SQLAlchemy, gRPC, GraphQL, RESTful API, WebSockets

**Databases & Search:** PostgreSQL, Oracle, Redis, Elasticsearch

**Cloud & Infrastructure:** AWS, Amazon EKS, AWS Lambda, AWS Fargate, Amazon S3, SageMaker

**DevOps & MLOps:** Docker, Kubernetes, Terraform, MLflow, GitHub Actions, CircleCI, Azure DevOps Pipelines

**Frontend & Visualization:** React, Next.js, Plotly Dash, D3.js, Amazon CloudFront

**Streaming & Messaging:** Apache Kafka

## EDUCATION

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**Bachelor's Degree in Computer Science**

Aug 2013 – May 2017

University of Texas at Austin, Austin, TX

## PROJECT

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**AI-powered Geospatial Intelligence Platform – Mapline.ai**

<https://mapline.ai>

AI-powered geospatial due diligence platform that leveraged LLMs, semantic search, and vector databases to automate analysis of zoning laws, environmental regulations, and geospatial constraints. Implemented a Retrieval-Augmented Generation (RAG) pipeline to transform fragmented municipal documents into structured, property-specific insights, enabling real-time decision support through a full-stack web application.

**AI-Powered Media Intelligence & Article Analysis Platform – 10Fold**

<https://10fold.com>

Developed an AI-driven NLP platform to automate large-scale analysis of online news articles for B2B PR and communications teams. The system ingested thousands of article URLs, applied NLP and semantic similarity models for deduplication, classification, and trend analysis, and generated structured, exportable reports through an intuitive dashboard. The platform eliminated manual article sorting, improved data accuracy, reduced analysis time, and enabled deeper insight extraction from high-volume media data.